

Project Synopsis

Title :

AI Mental Health Assistance System

Introduction :

Mental health issues such as stress, anxiety, depression, and overthinking are increasing rapidly, especially among students and young individuals. Many people hesitate to seek professional help due to stigma, lack of awareness, or limited access to mental health services.

This project proposes an AI-based Mental Health Assistance System that allows users to express their feelings through voice or text. The system analyzes the emotional state of the user using artificial intelligence techniques and provides supportive responses, along with personalized suggestions such as breathing exercises, music, and activities to improve mental well-being.

Problem statement :

Mental health problems often go unnoticed due to lack of early detection and reluctance to seek help. Existing solutions are mostly limited to simple chatbots and do not analyze emotions effectively using both voice and text.

There is a need for an intelligent system that can:

- Detect emotional states accurately
 - Provide real-time support
 - Offer personalized coping strategies
 - Track mental health over time
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Objectives :

The main objectives of this project are:

- To develop a system for detecting emotions from user input
- To analyze both voice and text data using AI techniques
- To design an empathetic chatbot for interaction
- To provide personalized mental health suggestions

- To track user mood and generate insights
 - To create a simple and user-friendly interface
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Proposed System :

The system works as follows:

1. User provides input through voice or text
 2. Voice input is analyzed for tone, pitch, and stress levels
 3. Text input is processed using NLP techniques
 4. Emotional state (stress, anxiety, depression, etc.) is detected
 5. Chatbot interacts with the user empathetically
 6. System suggests activities such as breathing exercises, music, or journaling
 7. User data is stored for mood tracking
 8. Dashboard displays emotional trends and insights
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Technologies Used :

- Python
- FastAPI / Flask
- React.js / HTML, CSS, JavaScript
- PostgreSQL / SQLite
- HuggingFace Transformers (BERT/Roberta)
- Wav2Vec2 (Voice Analysis)
- Librosa
- Claude API / OpenAI API
- Pandas & NumPy
- Pydantic

System Architecture :

Input → Voice/Text Input



Emotion Detection (NLP + Voice Analysis)



Processing & Classification



Chatbot Response



Suggestion Module (Activities/Music)



Database (Mood Storage)



Dashboard Output

Expected Output :

1. Accurate detection of emotional states using AI models
2. Real-time chatbot interaction for mental health support
3. Personalized suggestions based on user mood
4. Mood tracking and visualization over time
5. Improved emotional awareness among users
6. Early identification of stress and anxiety patterns